


```
[9]: ## print the shortest 10 script lines:
spark.sql("""
SELECT int(word_count) AS wc, raw_text
FROM lines_table
WHERE TRY_CAST(word_count AS INT) > 0
ORDER BY wc
""").show(10, False)
```

```
+-----+
|wc|raw_text|
+-----+
|1|Homer Simpson: Oh.|
|1|Homer Simpson: And?|
|1|Bart Simpson: Lewis?|
|1|Homer Simpson: (SHOCKED) Me?|
|1|Wendell Borton: Yyyyyyyyyyyyyyy!|
|1|Lisa Simpson: Baboon!|
|1|Lisa Simpson: Yeah.|
|1|Lisa Simpson: Nol.|
|1|Homer Simpson: Oh.|
|1|Homer Simpson: Nuts.|
+-----+
only showing top 10 rows
```

```
[11]: ## count the scenes that took more than 10 minutes
spark.sql("""
SELECT count(*)
FROM lines_table
WHERE TRY_CAST(timestamp_in_ms AS INT) > (10 * 60 * 1000)
""").show()
```

```
+-----+
|count(1)|
+-----+
|      86678|
+-----+
```

```
[13]: ## print the first 50 locations that has the word "Springfield" in them, ignoring letters case
spark.sql("""
SELECT DISTINCT raw_location_text
FROM lines_table
WHERE LOWER(raw_location_text) LIKE '%springfield%'
LIMIT 50
""").show(50, False)
```

[Stage 9:] (0 + 1) / 1

```
+-----+
|raw_location_text|
+-----+
|Springfield Police Station|
|Springfield YMCA|
|SPRINGFIELD REPUBLICAN PARTY CASTLE|
|Springfield|
|Springfield Pond|
|Springfield Mall|
|SPRINGFIELD ANIMAL REFUGE|
|Springfield Community Theater|
|SPRINGFIELD - THE PAST|
|PRIMORDIAL SPRINGFIELD|
|SPRINGFIELD HOSPITAL HALLWAY|
|SPRINGFIELD ELEMENTARY SCHOOL BUS|
|Springfield Elementary School Band Room|
|Springfield Opry House|
|Springfield Swap Meet|
|SPRINGFIELD PIER|
|Springfield Air and Space Museum|
|Springfield Trade Center|
|DREAMLIKE SUBURBAN SPRINGFIELD STREET|
|SPRINGFIELD MONUMENT PARK|
|Lake Springfield|
|SPRINGFIELD BATTING CAGE|
|Springfield Coast Highway|
|STREET NEXT TO SPRINGFIELD ELEMENTARY|
|SPRINGFIELD - EVENING|
|WINTRY SPRINGFIELD STREET|
|Springfield Wall of Fame|
|Springfield Shopper|
|Springfield Prison|
|Springfield Buddhist Temple|
|SPRINGFIELD STATE BUILDING|
|Springfield - Homer's Boyhood|
|SOARIN' OVER SPRINGFIELD RIDE|
|Springfield Public Beach|
|TASTE OF SPRINGFIELD|
|Springfield Pharmaceuticals|
|Springfield Elementary School Grounds|
|Cliff Overlooking Springfield Harbor|
|Springfield Mall Parking Lot|
|FIELDS AROUND SPRINGFIELD|
|SPRINGFIELD TIRE FIRE|
|PALM SPRINGFIELD|
|SPRINGFIELD PLAYGROUND|
|SPRINGFIELD - DOWNTOWN|
|SPRINGFIELD TV STORE|
|Springfield Minimum Security Prison|
|Springfield Aquarium|
|Springfield National Forest|
|SPRINGFIELD GYM|
|SPRINGFIELD MARINE PARK|
+-----+
```

```
[14]: ## Load the Locations dataset
df_locs = spark.read.option("header", "true").csv("simpsons/simpsons_locations.csv")

## create a view
df_locs.createOrReplaceTempView("locations_table")

## print 20 quotes that are located in any place that has Jerusalem in its name.
spark.sql("""
SELECT
    loc.name AS location_name,
    line.raw_text AS quote
FROM lines_table AS line
JOIN locations_table AS loc
ON line.location_id = loc.id
WHERE LOWER(loc.name) LIKE '%jerusalem%'
LIMIT 20
""").show(20, False)

+-----+
|location_name|quote|
+-----+
|JERUSALEM ROYAL PALACE|(JERUSALEM ROYAL PALACE: ext. jerusalem royal palace - plaza - day)|
|JERUSALEM ROYAL PALACE|Bart Simpson: I'm bored. Send in my jester.|
|JERUSALEM ROYAL PALACE|Krusty the Clown: Hey, hey, King David! (GOES INTO ROUTINE) Ha, ha, how ya doin'? Now I'm not saying Jezebel's easy, but befo|
|JERUSALEM ROYAL PALACE|Bart Simpson: What else you got?|
|JERUSALEM ROYAL PALACE|Krusty the Clown: Ahh... yeah... well, wait a minute, I got somethin' on the Canaanites. (MOAN) They are so stupid.|
|JERUSALEM ROYAL PALACE|Methuselah Grampa: (PAINED SOUNDS)|
|JERUSALEM ROYAL PALACE|Bart Simpson: Methuselah, my oldest friend. Who did this to you?|
|JERUSALEM ROYAL PALACE|Methuselah Grampa: (WEAK, PAINED GROAN) It was... Goliath.|
|JERUSALEM ROYAL PALACE|Bart Simpson: (PUZZLED) But Goliath is dead. I smote him myself. I smoted him good.|
|JERUSALEM ROYAL PALACE|Methuselah Grampa: No, it was his son, Goliath Two.|
|JERUSALEM ROYAL PALACE|Bart Simpson: (MELODRAMATIC) Nooo... (THEN, STEELY) Goliath Two is gonna pay. And this time it's biblical.|
|Jerusalem|(Jerusalem: EXT. JERUSALEM - FIELD OF BATTLE)|
|Jerusalem|Bart Simpson: (COCKY) I'll just give Goliath Two the ol' rock to the head, applause applause, "ya gotta believe in yourself,|
|Jerusalem|Santa's Little Helper: (DOPEY "DAVEY AND GOLIATH" VOICE) I don't know, Davey.|
|Jerusalem|Bart Simpson: (ANNOYED) Quiet, you.|
|Jerusalem|Santa's Little Helper: Oh, you've gotten pretty fat, Davey...|
|Jerusalem|Goliath/nelson: (OMINOUS) Let's get it on...|
|Jerusalem|Santa's Little Helper: See ya later, Davey.|
|Jerusalem|Bart Simpson: You killed my best friend.|
|Jerusalem|Crowd: (EXCITED "WOOO")|
+-----+
```

```
[15]: ## print first 20 most used Locations with the count of Lines spoken in them.
```

```
spark.sql("""
SELECT
    raw_location_text,
    COUNT(*) AS line_count
FROM lines_table
WHERE raw_location_text IS NOT NULL
GROUP BY raw_location_text
ORDER BY line_count DESC
LIMIT 20
""").show(20, False)
```

raw_location_text	line_count
Simpson Home	35059
Springfield Elementary School	7092
Moe's Tavern	4628
Springfield Nuclear Power Plant	3594
Kwik-E-Mart	1476
First Church of Springfield	1416
Simpson Living Room	1378
Springfield Street	1334
Springfield	1314
Simpson Car	1239
Flanders Home	1166
Street	1124
Springfield Town Hall	1103
Springfield Retirement Castle	1049
Burns Manor	998
Springfield Mall	833
Simpson Kitchen	816
Courtroom	813
Bart's Treehouse	767
Bart's Bedroom	766

```
[18]: ## find the seasons in which the average imdb rating was the highest.
## Print the seasons number, the number of episodes in each one and the average rating
## in a descending order from highest average rating to lowest.
```

```
## Load the episodes table
df_episodes = spark.read.option("header", "true").csv("simpsons/simpsons_episodes.csv")
df_episodes.createOrReplaceTempView("episodes_table")

## find the seasons in which the average imdb rating was the highest.
spark.sql("""
    SELECT
        season,
        COUNT(id) AS episode_count,
        AVG(CAST(imdb_rating AS FLOAT)) AS avg_rating
    FROM episodes_table
    GROUP BY season
    ORDER BY avg_rating DESC
""").show(35, False)
```

```
+-----+-----+-----+
|season|episode_count|avg_rating|
+-----+-----+-----+
|5|22|8.336363554000854|
|7|25|8.32400001525879|
|6|25|8.312000102996826|
|4|22|8.268181844191117|
|8|25|8.22|
|3|24|8.154166638851166|
|2|22|8.040909073569559|
|9|25|7.844000053405762|
|1|13|7.80769238105187|
|10|23|7.569565233976944|
|12|21|7.361904780069987|
|11|22|7.290909138592807|
|13|22|7.140909043225375|
|14|22|7.077272761951793|
|15|22|7.045454545454546|
|16|21|7.0428571701049805|
|18|22|7.000000043348833|
|19|20|6.934999942779541|
|20|21|6.895238058907645|
|17|22|6.863636341961947|
|25|22|6.831818125464699|
|21|23|6.821739093117092|
|23|22|6.813636324622414|
|22|22|6.8045454675501045|
|27|22|6.786363645033403|
|24|22|6.777272722937844|
|26|22|6.704545519568703|
|28|4|6.599999904632568|
+-----+-----+-----+
```

```
[21]: ##my question
##which 5 characters have the most dialogue lines in the series?
```

```
spark.sql("""
    SELECT
        raw_character_text,
        COUNT(*) AS total_lines
    FROM lines_table
    WHERE raw_character_text IS NOT NULL
    GROUP BY raw_character_text
    ORDER BY total_lines DESC
    LIMIT 5
""").show(5, False)
```

```
+-----+-----+
|raw_character_text|total_lines|
+-----+-----+
|Homer Simpson|29842|
|Marge Simpson|14159|
|Bart Simpson|13777|
|Lisa Simpson|11502|
|C. Montgomery Burns|3172|
+-----+-----+
```